

PROTOCOL FOR:

Spectral Recording of Gene Expression History by Fluorescent Timer Protein

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LEGEND

 **ATTENTION**

 **HINT**

 **REST**

REAGENTS AND MATERIALS

- The FT-ESARE/ArcMin plasmid can be obtained from the authors directly.
- HeLa cells (ATCC, CCL2)
- 6-well cell culture plates (Eppendorf, Germany)
- DMEM (Life Technologies, Germany)
- FCS (fetal calve serum, Life technologies, Germany)
- Pen/Strep (Penicillin/Streptomycin, Life Technologies, Germany)
- L-Glutamine (Life Technologies, Germany)
- TransIT-LT1 transfection reagent (MIR2304, Mirus Bio LLC, USA, WI)
- Opti-MEM I reduced serum medium (Life technologies, Germany)
- Phorbol-12-myristate 13-acetate (TPA, P8139, Sigma-Aldrich, Austria)

- PBS (Life Technologies, Germany)
- Trypsin/EDTA (Life Technologies, Germany)

PROCEDURE

We do not go into details regarding the plasmid cloning process, as we welcome everyone who is interested in using it, to contact us and we will provide it. Therefore, we will only describe the cell culture experiments.

Cell Culture and Transfection

1. HeLa cells are grown overnight in standard incubator conditions (37.5 ° C, 5 % CO₂) on 6-well plate in standard medium (DMEM + 10% FCS + 1%Pen/Strep + 1% L-Glutamine) to a confluency of about 80 %
2. Transfection with the TransIT-LT1 transfection reagent according to manufacturer: 1µg/µl plasmid in Opti-MEM I reduced serum medium, add TransIT-LT1 reagent and incubate for 30 min at RT, then add to cells for 24 h
3. Remove medium containing transfection reagent, wash with PBS and add fresh standard medium

FACS (alternative I)



4. Leave cells undisturbed until induction (we recommend 48 h)

5. baseline measurement: cells dislodged with trypsin and fluorescence and cell number determined by FACS



6. for induction with TPA: TPA is highly cancerogenic so wear respective protective gear!

Also, TPA is very volatile: make sure you only add TPA to the wells, in which you want to induce the cells!



7. Grow + and – TPA cells in separate 6-well plates to guarantee no TPA spills over into -TPA wells, as also the smallest amount of TPA induce IEG expression.

8. incubate for 40 min with 100ng/ml TPA diluted in standard medium

9. carefully remove all TPA and also remove medium of -TPA cells

10. wash with PBS

11. add fresh medium **or**

12. add about 2 ml trypsin until cells are detached from the plate
13. add the same amount of standard medium as trypsin to the cells for deactivation
14. pellet cells by centrifugation
15. resuspend in PBS
16. perform FACS measurements at the time points of your choice (we recommend every two hours for the first 9h and again at 24h post-induction)
17. FACS data can then be analyzed with any standard FACS analysis program (e.g. FlowJo)

Life-cell Imaging (alternative II)

4. 24 h after transfection cells are transferred into chambered cell culture slides

 *HINT*

5. Take cell growth into account when determining your cell density seeded onto the plates as they will stay on the culture slides for another 48 h

 *REST*

6. Let cells rest for 24 h in the incubator on the cell culture slides

 *ATTENTION*

7. for induction with TPA: TPA is highly cancerogenic so wear respective protective gear!

Also, TPA is very volatile: make sure you only add TPA to the wells, in which you want to induce the cells!

8. Add 100 ng/ml TPA diluted in standard cell culture medium

** HINT*

9. To avoid inducing – TPA cells with TPA, cover wells with parafilm while working with TPA. TPA is very volatile and the smallest amounts are enough to induce IEG expression.

10. Incubate cells for 40 min with TPA

11. Remove all medium (also from -TPA cells), wash cells with PBS and add fresh medium

12. Image cells with fluorescent microscope capable of live cell imaging over 24 h

13. We recommend imaging every hour

14. Images can then be analyzed with any software of choice (e.g. ImagePro Premier, ImageJ, ...)

15. For the calculations: a separate standard curve has to be performed for every independent experiment

16. When standard curve parameters are set, the cells of interest can be evaluated according to the mathematical approach described in detail in the manuscript

RECIPES

standard medium: DMEM + 10% FCS + 1%Pen/Strep + 1% L-Glutamine

EQUIPMENT

- Cell Culture working space (incubator, hood, centrifuge)
- FACS (BD LSR Fortessa2, BD Biosciences)
- Chambered cell culture slides (8-well Cell Culture Slides, CCS-8, MatTek Corporation)
- Spinning Disk microscope (Spinning Disk Confocal UltraView Vox, controlled by software Volocity)

TROUBLESHOOTING

If you see a positive signal in your – TPA cells, try handling your cells with even more care, as the FT expression can be activated very easily. Make sure medium is really on 37° C and not too cool before you add it. Avoid any type of stress for your cells as good as possible.